

ENM 5310: Data-driven Modeling and Probabilistic Scientific Computing

Lecture #1

Probability theory, statistics, and information theory fundamentals



Probability is the language of ML

Machine learning is often perceived as an art because people present it as a menu of methods.

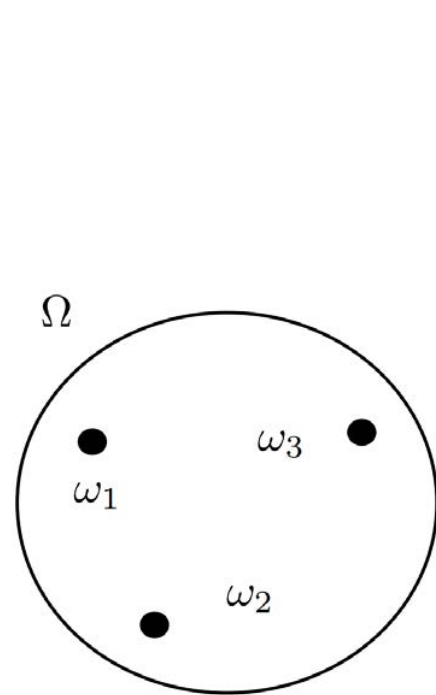
It is a science because every item on that menu is a probabilistic statement plus a decision rule — and once you see that, the menu collapses into a handful of fundamental ideas.

loss functions $\Leftrightarrow$ negative log-likelihood

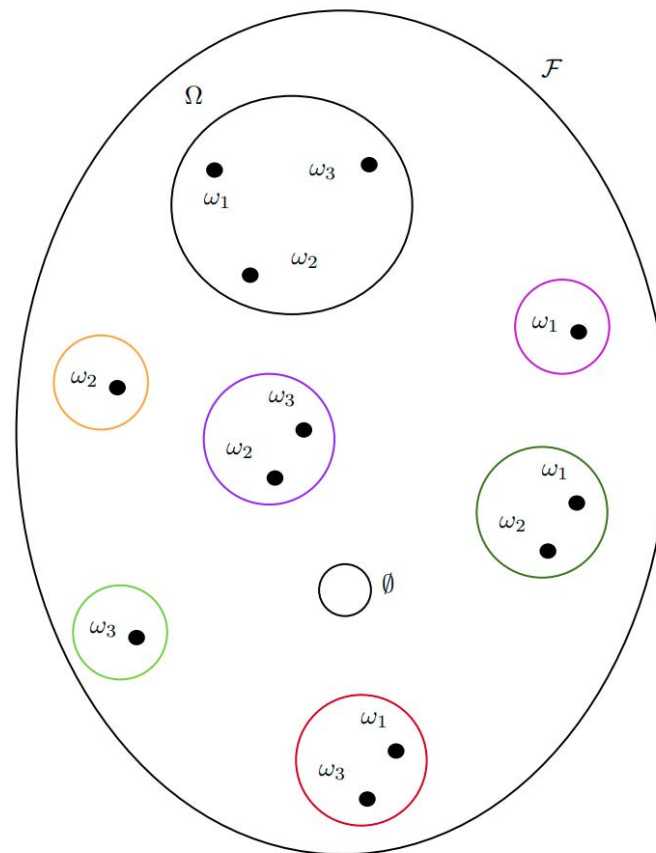
regularizers $\Leftrightarrow$ statistical priors

data/predictions $\Leftrightarrow$ statistical distributions

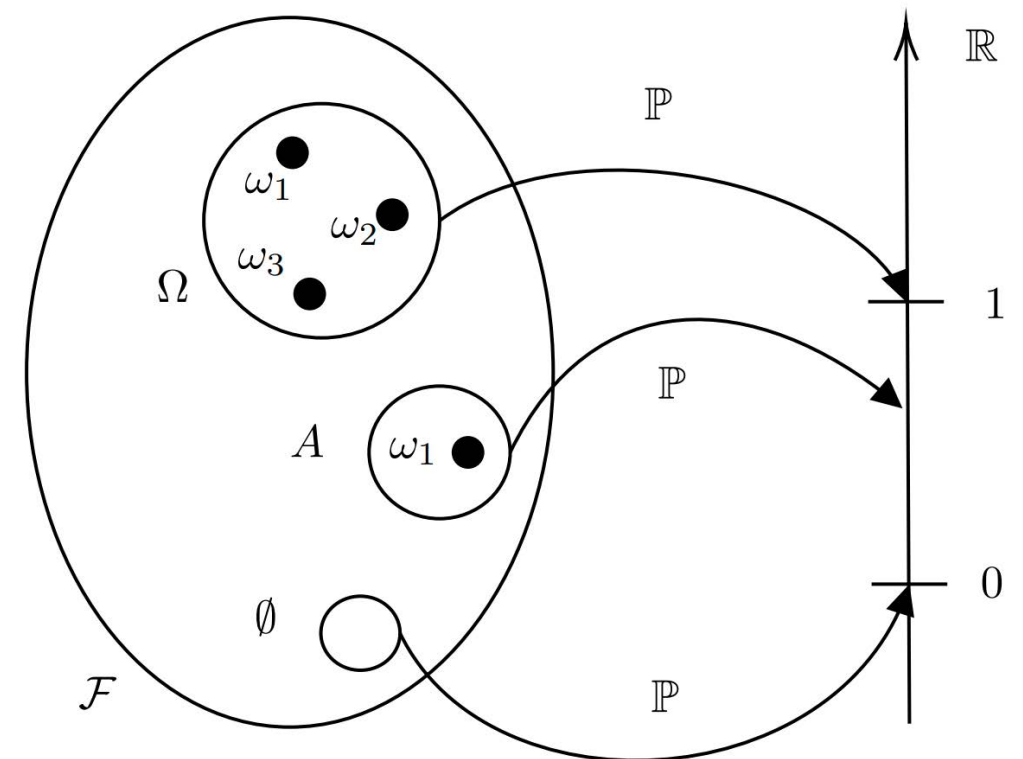
Probability spaces & random variables



Sample space



σ -algebra of events



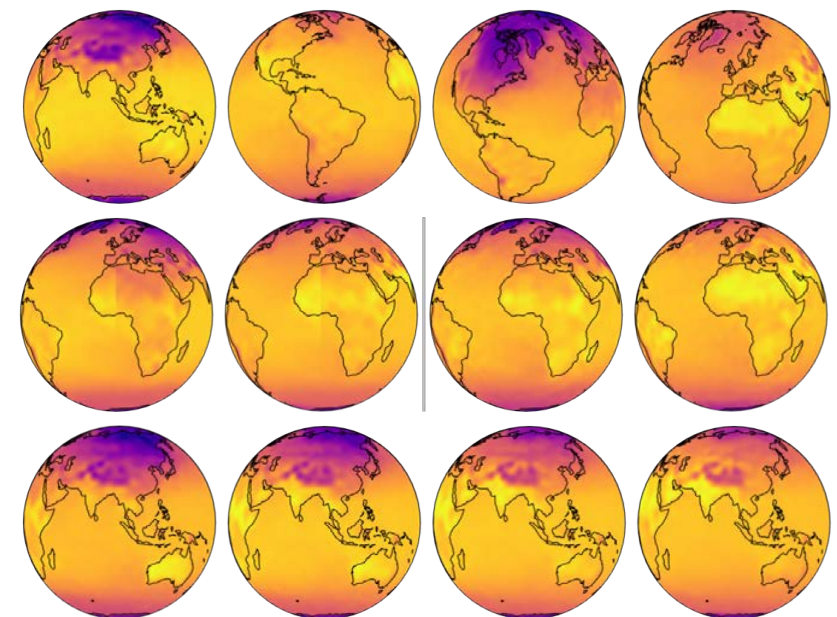
Probability measure



vectors



matrices



functions

Discrete random variables

- A ***discrete random variable*** is one which may take on only a countable number of distinct values such as 0,1,2,3,4,..... Discrete random variables are usually (but not necessarily) counts. If a random variable can take only a finite number of distinct values, then it must be discrete. Examples of discrete random variables include the number of children in a family, the Friday night attendance at a cinema, the number of patients in a doctor's surgery, the number of defective light bulbs in a box.
- The ***probability distribution*** of a discrete random variable is a list of probabilities associated with each of its possible values. It is also sometimes called the probability function or the probability mass function.

Continuous random variables

- A *continuous random variable* is one which takes an infinite number of possible values. Continuous random variables are usually measurements. Examples include height, weight, the amount of sugar in an orange, the time required to run a mile.
- A continuous random variable is not defined at specific values. Instead, it is defined over an *interval* of values, and is represented by the *area under a curve* (in advanced mathematics, this is known as an *integral*). The probability of observing any single value is equal to 0, since the number of values which may be assumed by the random variable is infinite.
 - Suppose a random variable X may take all values over an interval of real numbers. Then the probability that X is in the set of outcomes A , $P(A)$, is defined to be the area above A and under a curve. The curve, which represents a function $p(x)$, must satisfy the following:
 - **1:** *The curve has no negative values ($p(x) \geq 0$ for all x)*
 - **2:** *The total area under the curve is equal to 1.*
 - A curve meeting these requirements is known as a *density curve*.

Basic rules of probability

Sum rule $p(X) = \sum_Y p(X, Y)$

Product rule $p(X, Y) = p(Y|X)p(X)$

Bayes rule $p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$